

# Controlled Error-Geometry Experiment on GBnetwork

Fixed-norm directional intervention across PIGNN-GC, GridSFM, GridFM-GraphKit, and LUMINA

## 1 Question and conclusion

The question is whether reconstructed AC power-balance inconsistency is more sensitive to the direction of voltage-magnitude error outside a training-derived solution subspace than to error within it. The experiment holds the error norm fixed separately for every held-out scenario, retains the model’s calibrated phase angle, and changes only the magnitude-error direction.

All four 40-epoch surrogate families exhibit the same directional contrast: removing the off-subspace component reduces mean power balance (PB), whereas removing the in-subspace component increases it. The effect is largest for PIGNN-GC and GridSFM, intermediate for GraphKit, and small but statistically nonzero for LUMINA. This is a diagnostic counterfactual that uses the Newton reference state; it is *not* a deployable correction. The separate CSP rows below are the practical frozen-output intervention.

## 2 Frozen protocol

GBnetwork Backbone v2 contains 37,022 converged 2224-bus AC power-flow scenarios. The split is `random_split` with seed 42: 12,339 train, 12,339 validation, and 12,344 test scenarios. The four inputs are the best-validation checkpoints from the matched 40-epoch campaign. Per-bus magnitude offsets and circular angle offsets are fitted on training predictions only and frozen before validation or test scoring.

For predicted and reference calibrated magnitudes, let

$$e = \hat{v}^C - v^*, \quad e^\parallel = P_k e, \quad e^\perp = (I - P_k)e,$$

where  $P_k$  is fitted only to centered training reference magnitudes. At each test scenario and  $\alpha \in \{0, .25, .5, .75, 1\}$ , the two counterfactuals are

$$\tilde{e}_\perp(\alpha) = \|e\|_2 \frac{e^\parallel + \alpha e^\perp}{\|e^\parallel + \alpha e^\perp\|_2}, \quad \tilde{e}_\parallel(\alpha) = \|e\|_2 \frac{\alpha e^\parallel + e^\perp}{\|\alpha e^\parallel + e^\perp\|_2}.$$

No clipping or epsilon approximation was used; a zero original error maps to a zero counterfactual, and a nonzero error with a zero denominator would fail the run. The scorer recomputes complex128 AC residuals from the modified complex voltage. Its GENCO-compatible per-bus score is

$$r_i^{\text{PB}} = \begin{cases} \sqrt{\Delta P_i^2 + \Delta Q_i^2}, & i \in \text{PQ}, \\ |\Delta P_i|, & i \in \text{PV}, \\ 0, & i \in \text{slack}, \end{cases} \quad \text{Mean PB} = \frac{1}{SN} \sum_{s,i} r_{is}^{\text{PB}}.$$

## Validation-only rank selection

Candidate ranks were  $\{4, 8, 16, 32, 64\}$ . A rank was admissible only if CSP did not worsen validation  $|V|$  RMSE for *any* model. Among admissible ranks, the pre-declared objective minimized the mean of the four normalized PB values  $q_m(k) = \text{PB}_m^{\text{CSP}}(k)/\text{PB}_m^C$ . Every candidate met the RMSE guardrail;  $k^* = 16$  was selected before the test split was scored.

| Rank $k$                       | 4       | 8       | <b>16</b>      | 32      | 64      |
|--------------------------------|---------|---------|----------------|---------|---------|
| Mean $q_m(k)$ over four models | .272423 | .272030 | <b>.270474</b> | .271460 | .271975 |

## 3 Test results at frozen $k^* = 16$

Table 1 distinguishes the calibrated model state from its practical calibrated solution-subspace projection (CSP). CSP improves both the magnitude RMSE and Mean PB for all four families; the magnitude of this practical gain is architecture dependent.

Table 1: Frozen-output intervention on the 12,344-scenario test split. C is train-only calibration; CSP is C followed by  $P_{16}$ .

| Model           | Mean PB (C) | Mean PB (CSP) | $ V $ RMSE (C) | $ V $ RMSE (CSP) |
|-----------------|-------------|---------------|----------------|------------------|
| PIGNN-GC        | .14376      | .05690        | .01422         | .01289           |
| GridSFM         | .08660      | .05009        | .01179         | .00857           |
| GridFM-GraphKit | 1.45564     | .07801        | .02090         | .02026           |
| LUMINA          | 1.83683     | .10291        | .02331         | .02326           |

Figure 1 is the controlled mechanism experiment. Every curve is normalized by that model’s calibrated Mean PB at  $\alpha = 1$ ; hence all curves start at one and do not compare raw model quality. Off-subspace suppression reduces PB monotonically for PIGNN-GC and GridSFM. It also reduces PB for GraphKit and LUMINA, but only modestly. The in-subspace control moves in the opposite direction for every model.

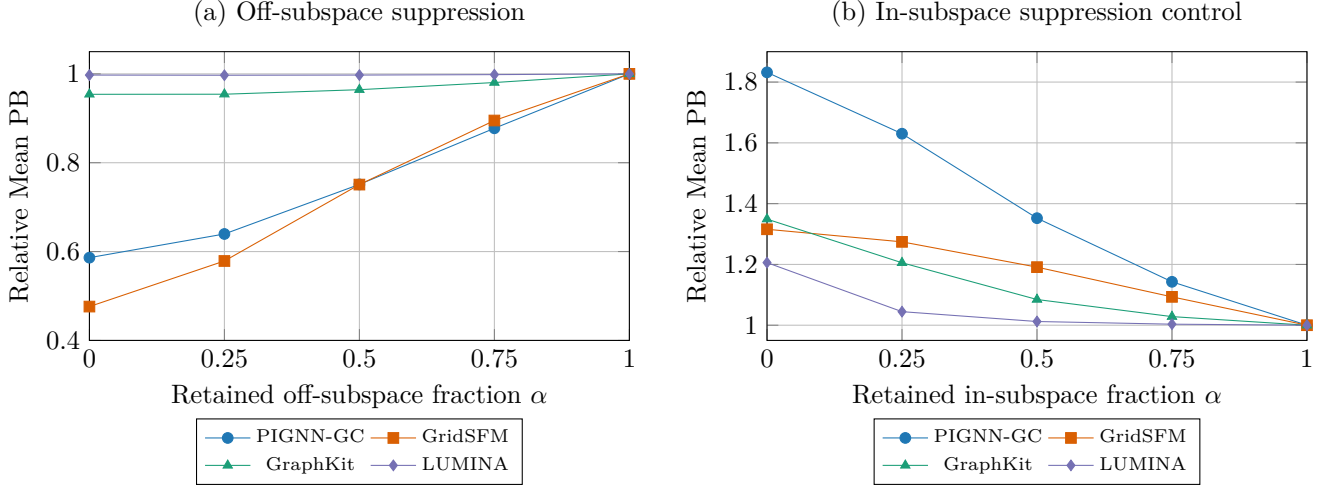


Figure 1: Fixed-norm directional diagnostic at  $k^* = 16$ . Each point changes only the orientation of the calibrated magnitude error and retains the calibrated phase angle. Lower is better.

Table 2: Endpoint robustness at  $\alpha = 0$ .  $\Delta$ PB is relative to  $\alpha = 1$ ; CIs are 95% paired bootstrap intervals over 12,344 scenarios.

| Model    | Off $\Delta$ PB [95% CI]   | PB lower (off) | In-control $\Delta$ PB [95% CI] |
|----------|----------------------------|----------------|---------------------------------|
| PIGNN-GC | -.05947 [-.05991, -.05904] | 100.000%       | +.11962 [+.11806, +.12118]      |
| GridSFM  | -.04536 [-.04569, -.04507] | 100.000%       | +.02734 [+.02706, +.02763]      |
| GraphKit | -.06690 [-.06715, -.06665] | 99.992%        | +.50814 [+.50227, +.51417]      |
| LUMINA   | -.00497 [-.00503, -.00490] | 99.174%        | +.37820 [+.37265, +.38403]      |

## 4 Interpretation and limitations

The experiment supports a directional explanation beyond total voltage-error magnitude: rotating the same error norm away from  $e^\perp$  is associated with lower AC PB, while suppressing  $e^\parallel$  is harmful. It does not identify the empirical SVD subspace with the AC-feasible set, does not prove that every off-subspace direction is physically invalid, and does not establish a causal architecture ranking. GraphKit and LUMINA show a much weaker off-subspace effect than PIGNN-GC and GridSFM; this heterogeneity should be retained in any paper figure or claim. The subspace is fitted to the training operating distribution, so this report does not establish out-of-distribution transfer.

## 5 Reproducibility record

The Stage-1 validation JSONs, frozen rank decision, Stage-2 summaries, and per-scenario PB arrays are saved under `results/controlled_geometry_20260905/`. Stage 2 used the frozen common rank only after the validation decision was recorded. The largest absolute fixed-norm error was  $8.88 \times 10^{-16}$ , the largest  $\alpha = 1$  PB reproduction error was exactly zero, and no zero-denominator or invalid case occurred. These checks rule out clipping, norm drift, or a mismatch between the counterfactual’s  $\alpha = 1$  endpoint and the calibrated prediction.